Who Gets a Place When Shortlists Get Better?

Evidence from New York rental simulations and Chilean school admissions

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A shortlist is scored one applicant at a time, but a seat or an apartment can go to only so many people. We study the consequences of that mismatch in two settings chosen for complementary evidence. New York rental advertisements and household records let us vary affordability and screening in a simulated market. Chilean school admissions supply submitted rankings, tie-break orders and recorded offers. We apply the same rule to five discretionary slots in both settings, retaining school continuity rights. Keeping applicants' highest-ranked options increases coverage of their full-list partner relative to random lists, yet reduces the number placed in both rental designs and both school cohorts tested with this benchmark. We then evaluate the recommenders specific to each setting. A rental language model shows more full-list partners but places fewer renters. Adding reported rank to a school offer predictor covers more recorded offers and places fewer pupils in each of four subsequent cohorts. Across 2,044,649 application records it covers 5,552 additional historical offers and yields 1,787 fewer placements in replay. Of 2,527 applicants who lost a place between the two school policies, 1,752 received exactly the same list. We characterize the largest set of menu changes that preserves a baseline stable assignment, and measure how much prediction accuracy remains available under that requirement. All alternative-policy placements are counterfactual.

1 Introduction

A family opens a housing search page or a school application and has room to consider five options. The system can give that family five good recommendations. It can also give thousands of other families those same five recommendations. An audit that scores each list against the family's eventual offer sees the first decision. To see the second, it must clear all the lists against the same supply.

We began with New York rentals because there is a prior question to ask before worrying about information or competition: does this household have an apartment it can afford at all? Advertisements give asking prices and bedrooms, and survey households give budgets and space needs. Together they let us hold an inventory fixed or deliberately make affordable options more common. The missing evidence is consequential. The survey and advertisements do not link a household to an application, an offer or a lease. We must therefore model both rankings. The centralized rental allocation in this paper is a modeling device. Real apartment searches involve applications to individual landlords; our replay imposes deferred acceptance to study competition on the displayed inventory.

We chose Chilean school admissions to test the part of the argument most exposed to that modeling decision. Families actually submit ordered applications, schools have limited seats, and the ministry records initial offers. These records let us ask whether a policy can retain more options that households demonstrably pursued while leaving fewer people placed under a replay of the assignment rule. Chile cannot validate the housing preference model. Conversely, school lists reveal little about opportunities families never considered. Pairing the settings gives us one setting in which we can alter feasible supply and a second in which we can observe the target used to score a shortlist. We first compare identical rules for choosing five discretionary options, preserving the entitlement that some Chilean pupils hold at their current school. We then compare the recommenders specific to each setting.

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We find a common direction, with very different magnitudes. Keeping the applicant's first five choices in place of five random choices covers more full-list partners and places fewer applicants in both rental designs and both original school cohorts. The tradeoff also appears between two predictors fitted to Chilean data. It does not appear for every comparison: both school predictors beat uniform shortlists on both outcomes in the 2022–23 benchmark. The choice facing a platform is which policy to replace.

The technical question is whether a platform could keep the useful changes and avoid the displacements. Given two sets of menus and a baseline stable allocation, we characterize exactly which individual switches can be combined while keeping that allocation stable. The calculation uses each applicant's old assignment, the newly visible pairs and program priorities. We run the resulting rule in both markets. Chile's continuity entitlement also lets us examine an institutional exception to the protection implied by a pure first-five list. We also study a different option available after tentative offers: a coordinated referral to a vacant, acceptable place. Both interventions have clear limits when too few households can use the available supply.

2 What each experiment measures

Let E be the bipartite graph of acceptable applicant–program pairs. A displayed graph $D \subseteq E$ lists the programs an applicant can pursue. Each applicant strictly orders acceptable programs above being unplaced. A program has a fixed capacity and a strict priority ordering over applicants. It holds the highest-priority acceptable applicants who have proposed, up to its capacity. In the rental simulations each listing has one place. Let $M = \mathrm{DA}(E)$ be the full-list allocation and $N_D = \mathrm{DA}(D)$ the policy allocation; an unplaced applicant receives $\perp$. Stability means that no displayed pair would both prefer to match, taking account of vacancies and priorities. DA on a fixed graph is polynomial-time and applicant-optimal among its stable allocations [Gale and Shapley, 1962, Roth, 1986].

For applicant i, define

$$C_i(D) = \mathbf{1}\{M(i) \neq \perp,\ (i, M(i)) \in D\}, \qquad P_i(D) = \mathbf{1}\{N_D(i) \neq \perp\}.$$

Both rates use all applicants. A person without a full-list partner contributes zero to coverage and can still receive a place after the lists change. In Chile we also compute C_i using the historical offer instead of $M(i)$. For policies A and B, the ordered pair $(C_i(B) - C_i(A), P_i(B) - P_i(A))$ belongs to a three-by-three table. The table identifies a person who newly sees their partner yet loses a placement. We use a separate indicator for exact equality of the two personal menus to identify people whose options stayed fixed while competitors' lists changed. Unchanged coverage by itself does not imply that a menu stayed fixed.

Relation to prior work. Competition between recommended options has a long history. Su et al. [2022] optimize rankings jointly when job seekers apply and employers decide whom to interview. Ashlagi et al. [2020] study recommendations that reduce communication in matching markets. McCauley et al. [2026] link unmatched hospitals under prefix truncation to missed stable partners. Our comparisons permit deletions anywhere on a submitted list and trace who is displaced. We use the classical invariance of matched participants on a fixed graph and results on entry and consistency [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986, Sasaki and Toda, 1992].

Aguiar and Hong [2026] compare stable and maximum assignments on restricted graphs and use Chilean school data to study retained rankings. We share their distinction between feasible and stable placements. We compare alternative fitted rules at a fixed quota and identify whose placement changes without a change in their own menu. Arteaga et al. [2022] show in a field intervention that advice about admission risk can induce families to add options and improve

Table 1. Two policy comparisons with a common allocation outcome. Each school application cycle is a separate national cohort.

	New York rentals	Chilean school admissions
Choices and priorities	Affordability and space rules, modeled tastes and screening	Submitted course rankings, published priorities and within-course tie orders
Shortlist change	Uniform five to Haiku five	Rank-blind predictor to rank-informed predictor, five discretionary choices plus entitlement
Reference option	Simulated full-list DA unit	Full-list replay program and recorded initial offer
Evidence	Development simulation, 20 seed blocks per supply design	Frozen 2019 models; two pairs of later cohorts for the specified signs

actual placements. Here all comparisons hold the submitted reports fixed. Munro [2026] studies how individual treatments spill over in designed markets, including Chilean school choice. We directly rerun specified menu policies on entire fixed cohorts and ask which changes preserve the original allocation. Liu et al. [2024] audit the housing options language models recommend. Our housing exercise follows each recommendation through competition for a particular unit.

3 What does showing a partner guarantee?

The full-list partner in this section is an assignment in the same model as the restricted replay. The Chilean historical offer provides a separate empirical target.

The best prediction can send two people to one seat. Three applicants share three one-seat programs x, y, z. Their first choices, in applicant order, are (x, y, z), (x, z, y), (y, x, z) and (y, z, x) with probabilities $2/6, 2/6, 1/6, 1/6$. Each state has distinct first choices, so all three receive those choices under full-list DA whatever the program priorities. The platform knows the probabilities but cannot see the state. With one slot per applicant, the unique choices maximizing expected partner coverage are (x, z, z). They cover $5/3$ partners in expectation and place two people. The lists (x, y, z) cover $3/2$ partners in expectation and place all three. Thus filling the third seat can cost as little as $1/6$ of an expected correct recommendation in this example.

What an omitted partner can explain. Let H be the applicants whose full-list assignment is a real program missing from D. Let K_j and T_j count members of H assigned to j under M and N_D, respectively.

Theorem 3.1. *Relative to the full market, gross losses and gross gains in placement on D are each at most $|H| - \sum_j \min(K_j, T_j)$. Furthermore, every unmatched applicant whose full-list partner is displayed can be paired with a distinct member of H who is placed on D and strictly prefers their full-list partner. Both conclusions hold for arbitrary stable selections on the two graphs.*

Proof. Delete H together with the seats its members occupy in each assignment. Any remaining blocking pair would have blocked the original assignment. The remaining applicants thus have stable outcomes on the same displayed pairs, using capacities $c - K$ and $c - T$, where c is the original capacity vector. In the first market their full-list partners all remain visible. For fixed pairs, adding one seat retains every matched applicant and adds at most one: regard seats as unit copies with common order within each program, run seat-proposing DA, and admit the new proposer last. An applicant who holds an offer can only trade up. Matched-set invariance transfers this comparison to any stable selection [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986]. Comparing the

two capacity vectors through their coordinatewise minimum bounds losses among the remaining applicants by $\sum_j (T_j - K_j)_+$ and gains by $\sum_j (K_j - T_j)_+$. Exactly $|H| - \sum_j T_j$ of H lose a place, and none gain. Adding gives the bounds.

For the individual claim, at each program cancel the holders common to both assignments. Pair departing holders with incoming holders one-to-one until one side is exhausted. Start with an applicant who is unmatched on D despite seeing their full-list partner j. Program j must be full in N_D, and every new holder there outranks this applicant. Follow the pairing at j to one of its new holders h. Full-list stability implies that h had an assignment strictly preferred to j, since j previously held a lower-priority applicant. If h still sees that former assignment, repeat. Each step starts with someone who strictly prefers their former assignment and each pairing is injective. On a continued step the program is full after restriction. Its intake of new holders is therefore at least its number of departing holders, so the departing applicant always has a successor. The starting applicant has no incoming link, so the paths cannot cycle or meet. Each path ends at a placed applicant whose former assignment is hidden. □

For two policies A, B, one entry in the empirical comparison counts applicants with $C_i(B) - C_i(A) = 1$ and $P_i(B) - P_i(A) = -1$. The theorem gives each such applicant a distinct other person on the final display who is assigned a worse program than their full-list partner, which is hidden under B. That other applicant can have the same list under A and B. It can happen with only two applicants and one slot each. Let i prefer x to y, let h prefer y to x, and let both programs give h higher priority. Full-list partners are x for i and y for h. Display A offers i only y and h only x. Both place. Change only i's list to x. Now i sees their full-list partner and loses the competition for it. Under one-slot lists every witness is this direct: a new holder at the focal program could not also see their own strictly preferred full-list partner. The traced links in a larger market describe the two assignments and need not follow the chronology of DA proposals.

Let $p_{ij}(X)$ be the probability that j is the full-list partner of i, given the information X used to choose the lists. If a policy also uses randomness independent of the full allocation conditional on X, condition on its seed as well. By linearity, the conditional expected number covered is $\sum_i \sum_{j \in D(i)} p_{ij}(X)$. Selecting the most probable partners separately for each person therefore minimizes expected omissions, the coarse upper bound $|H|$ above. A smaller upper bound can coexist with fewer actual placements. For comparison, McCauley et al. [2026, Theorem 4.7] identify which hospitals can become unmatched under one-sided prefix pruning. Here omissions can occur anywhere in an incomplete list, and a displaced person can still see their own full-list partner.

4 The same five-choice rule in both markets

Before comparing recommenders, we ask a question that has exactly the same meaning in the two settings. Suppose each person can list five options. What happens if everyone lists their five favorites? The alternative is to sample up to five feasible options uniformly without replacement for each applicant. In either case applicants propose in their original order. We keep any protected school entitlement outside the five slots. An applicant with at most five discretionary options sees them all. The benchmark uses modeled preferences in housing and families' submitted rankings in school choice. It tests what concentrating choices near the top of people's rankings can do at fixed capacity. The rental language model does not observe those realized rankings.

Choosing favorites raised full-list partner coverage and lowered placement in all four panels. To compute expected rental coverage under sampling, each full-list recipient with degree d contributes $\min(5, d)/d$. Placement requires running DA. We averaged 48 uniform draws per market; Monte Carlo standard errors on the estimated placement rates were at most 0.022 percentage points in NYC and 0.0012 in Chile. These standard errors describe the numerical uncertainty in averaging menus.

Table 2. First five minus uniform five. Both outcomes use all applicants as the denominator. Changes are percentage points. NYC rates use the study's geographic weights. School rates give each application record equal weight.

Panel	Applicant instances	Full-list partner shown	Placed
NYC ordinary	15,360	+21.03	−13.86
NYC enriched	15,360	+17.03	−18.88
Chile 2022	570,891	+0.82	−0.23
Chile 2023	536,353	+0.99	−0.23

They do not measure uncertainty about the housing behavioral assumptions or about future cohorts. Short lists and continuity protections in Chile mean that many pupils receive exactly the same choices under either rule. The larger rental differences do not show that a housing recommender is intrinsically more harmful. The shared direction shows that the reversal can arise under a common rule in both settings. The remaining comparisons evaluate what the actual candidate policies would change.

When keeping a favorite protects its recipient. There is a precise reason to distinguish the school entitlement from an ordinary five-choice cap. If every list is a prefix of the applicant's full order, an applicant whose full-list DA partner survives that cut is placed at that program or one they prefer. To see this, run applicant-proposing DA on the prefixes, pausing anyone who exhausts their prefix. Continue that same run with the omitted suffixes restored. Those already holding an offer can only keep it or propose farther down their lists. Someone unplaced at the pause has already had every prefix option rejected. Order independence of DA gives the full-list allocation at the end [Gale and Shapley, 1962, Gale and Sotomayor, 1985]. This is consistent with fewer placements relative to random menus: the guarantee concerns people whose full-list partner the prefix retains.

In the two rental designs every one of the 4,531 and 4,274 covered full-list recipients was indeed placed under first-five menus. In Chile an entitlement kept below the first five can break the prefix property. If an applicant ends up with a worse program while their full-list partner is hidden, their assignment must be one of those retained entitlements: a displayed top-five option could not be worse than an omitted one. The distinct witnesses from Theorem 3.1 must therefore end at such a returning pupil. We verified this for all 126 covered-but-unplaced applicants in 2022 and all 120 in 2023. The school institution supplies a concrete boundary of the guarantee.

5 New York City rentals

We paired archived model recommendations with a simulation of who could rent each apartment. The inputs are recent-mover household records from the 2020–24 American Community Survey and rental advertisements first listed in June–August 2026 [FirstMover, 2026, U.S. Census Bureau, 2026]. Each market contains 64 renter instances and 64 available apartments from one borough and month. An apartment must meet bedroom needs and cost at most 40% of monthly income. Lower-income households earn at most the survey-weighted median for NYC cash renters. Renter preferences combine rent, size, physical features, distance and taste shocks. Listing priorities depend on income relative to rent and a random shock specific to the renter–listing pair. We rerun applicant-proposing DA with exactly the same households, apartments and preferences under each display. The resulting placements are simulated. The ads provide asking rents, and the survey describes renter households. Neither source observes offers or leases between these parties.

In the ordinary design we sample households using survey weights and draw available apartments uniformly. In the enriched design half the inventory comes from a borough-specific affordability

Table 3. NYC. Change under Haiku relative to uniform five-listing menus, in percentage points, with paired 95% simulation intervals. Both columns use every renter in the row. A renter without a full-list DA partner contributes zero to partner coverage in both policies.

Design	Renters	Full-list partner shown	Placed by DA
Ordinary	All	+5.62 [4.96, 6.27]	−2.71 [−3.19, −2.23]
	Lower income	+0.11 [0.00, 0.24]	+0.00 [−0.19, 0.21]
	Higher income	+9.17 [7.99, 10.34]	−4.58 [−5.27, −3.86]
Enriched	All	+5.85 [5.04, 6.60]	−5.40 [−6.05, −4.73]
	Lower income	+0.78 [0.19, 1.43]	−0.92 [−2.23, 0.37]
	Higher income	+10.92 [9.47, 12.29]	−9.88 [−10.43, −9.36]

band. We then sample 32 lower-income households with a feasible option in that inventory and 32 higher-income households. This deliberate change lets us examine competition where lower-income households can participate. Each design has 20 seed blocks across three months and four supported boroughs, giving 240 paired markets and 15,360 renter instances. Staten Island had too few listings for draws of 64. The supported boroughs represent 97.95% of the intended recent-mover geographic weight.

The income cutoff is $64,435 in 2024 dollars. In asking-rent dollars at the study's July 2026 price base, the highest monthly rent allowed at the cutoff is $2,286. We use the calibrated preference specification from the rental pilot. It selects a rent coefficient of 4 and an idiosyncratic shock scale of 0.5 from a nine-point grid. Its burden calibration has an observable limitation. At 217 lower-income placements across the 240 ordinary calibration markets, even a feasible assignment chosen to minimize their first-quartile rent burden cannot push that quartile below 27.78%. The target quartile from recent movers' contract rents was 22.00%. These finite-market bounds concern the calibration exercise. The household survey does not validate the assumed listing priorities or individual rankings.

What the recommender sees. For each renter, the five-slot uniform rule draws from the feasible listings without replacement. The archived Claude Haiku 5.5 recommender receives the same candidate set, a budget, bedroom needs, target area, prices, physical features and distances. The prompt asks it to choose five listings. It has no observations of other renters' menus or the realized preferences. Both policies show all feasible options to anyone with five or fewer. We used the 20,740 saved valid responses from the development pilot. Households and apartments can recur across draws. Rates below pool individual indicators within each borough-month and then average cells with equal month weights and Census recent-mover borough weights. The pointwise intervals resample entire paired seed blocks 10,000 times. They condition on the catalogues, behavioral model and stored replies.

What changed when everyone used the model? Under the original pilot design the primary endpoint was the lower-minus-higher difference in hidden blocking pairs. The comparison here follows the same people and uses the saved menus. Ordinary markets placed 53.89% of renters under uniform lists and 51.18% under Haiku, even as full-list partner coverage rose from 9.42% to 15.04% (Table 3). The direction depends on the comparator. Against a fixed score that ranks eligible apartments using their observable features, Haiku placed 20.26 points more renters in ordinary markets and 24.47 points more in enriched markets. Its coverage differences were −0.44 and +0.66 points, respectively.

What the group averages conceal. In the ordinary design, 5,425 of 6,617 lower-income renter instances had no feasible listing. The standardized change in their placement rate was smaller

than one hundredth of a point. In enriched markets the change was −0.92 points with an interval spanning zero, yet 1,203 lower-income instances lost a placement and 1,044 gained one. The enriched lower-income share with a hidden blocking pair (a renter and an unshown apartment that each prefer the other to their assignment) rose from 37.03% to 41.31%. The higher-income share fell from 85.19% to 78.01%. The absolute gap narrowed even as hidden blocking pairs became more common for the lower-income group. Partner coverage and placement were secondary outcomes in this development pilot. We treat these results as evidence of a mechanism under specified assumptions.

Which housing conclusions depend on the preference model? We kept the same households, inventories, feasibility rules, and saved recommendations, then changed one assumption at a time. We used income-blind listing lotteries, a common priority by income, a smaller rent penalty (1 instead of 4), and larger idiosyncratic taste shocks (scale 1.25 instead of 0.5). Each scenario recomputes full-list partners and assignments. In all four, Haiku improved aggregate coverage and lowered aggregate placement in both supply designs. The lower-income result was more sensitive. In enriched markets the estimated placement change was −0.92 points under the calibrated model, +2.19 [1.13, 3.22] under common income priority, and −2.73 [−3.82, −1.61] under income-blind priorities. The ads and survey do not tell us which priority specification best describes real landlords. This sign change limits the distributional claim the simulations can support.

6 Chilean school admissions

The Chilean school choice system records the courses a family ranks, the school's priority and lottery information, and the initial offer the family actually receives [Correa et al., 2022, Ministerio de Educación de Chile, 2026]. We can ask whether a hypothetical shortlist would have included that offer. To learn who would receive a place on the new lists, we must replay the allocation.

We fitted two ridge logistic offer predictors to the same 483,070 applicants in 2019, using each eligible applicant–course edge as an observation. Both use program demand, capacity, congestion, list length, grade, priority indicators and a smoothed program offer rate estimated from 2019 (leave-one-out for training). The second adds log rank, a first-choice indicator and a rank–congestion interaction. Neither policy sees the realized offer. The lottery is used for allocation only. Each predictor retains up to five discretionary choices. A pupil's guaranteed current-school option remains available outside that quota. The selected courses keep their submitted order when applicants propose. We use the same reported priorities, realized within-course tie orders and pooled capacities for both replays. Full-list DA reproduces the recorded program or outside option for 95.39% of applicants in 2022 and 95.79% in 2023.

The first confirmation and its extension. After examining the 2020 and 2021 application years, we recorded a joint prediction before obtaining the 2022 and 2023 releases. These application years correspond to school entry in 2023 and 2024. The cohorts include 520,420 and 492,888 initial offer recipients, respectively. All applicants enter the placement and joint-transition counts. Giving the 2019 predictor access to rank would cover more historical initial offers and reduce replayed placements in each subsequent cohort. Table 4 reports both outcomes, with full-list DA partners as a second, directly comparable coverage target. Among the 2,493 net additional historical offers covered in 2022–23, 2,244 were offers that families explicitly accepted. The historical record does not tell us whether they would have accepted under either new policy.

Before downloading either of the next two releases, we committed the five-slot comparison, the two directional criteria and the inclusion of both 2024 and 2025 application years. We applied the same 2019 coefficients, 2019 program frequencies, quota, capacity adjustments and priority ordering without refitting. The original models passed both criteria in each additional year. Across these two new cohorts of 937,405 application records the rank-informed predictor covered 3,059 more

recorded offers and placed 802 fewer applicants. Among the 1,072 who lost a placement, 748 had identical personal menus under both policies. Full-list replay agreed with the recorded program or outside option for 96.59% in 2024 and 96.51% in 2025. We reproduced every pupil's assignment with two implementations of DA under both policies and full lists. The fixed-priority model uses the within-course order in each release. For school entry in 2026 (application year 2025), the ministry replaced a random tie-break with a deterministic applicant–school ordering; the published field still supplies the realized order [Ministerio de Educación de Chile, 2025, 2026].

Table 4. Rank-informed minus rank-blind school menus. Placement and full-list DA coverage come from the replay. Historical offer coverage uses the recorded initial course. Each entry is a net count in the same application-year cohort.

Year	Applicants	Historical offers shown	Full-list partners shown	Placements
2022	570,891	+1, 143	+1, 261	−468
2023	536,353	+1, 350	+1, 473	−517
2024	473,482	+1, 735	+1, 841	−481
2025	463,923	+1, 324	+1, 400	−321

The choice of comparison matters. In the 2022 and 2023 cohorts we also reran DA on 48 uniform five-choice displays in each cohort and on the family's first five choices. Relative to the uniform-draw mean, the rank-informed policy covered about 9,235 and 9,740 more full-list partners and placed about 789 and 759 more pupils in 2022 and 2023. Relative to first-five, it placed 2,123 and 2,013 more. The rank-blind predictor placed the most pupils of these four policies in both years. Prioritizing predicted offers improved on simple lists, and adding rank information then moved the two outcomes in opposite directions.

Pooling all seats simplifies the ministry's allocation. A sensitivity that first honors continuity and gives designated pupils priority for earmarked vacancies changes the year-specific net placement losses to 469 and 516. Neither version models all reserves or dynamic family priorities. These are whole-cohort comparisons under fixed reports and the published within-course tie orders. The person-level transitions and comparisons with uniform are exploratory. The joint directional test involved only the recorded-offer and placement outcomes under the two frozen fitted policies.

List length. Keeping both fitted scores fixed, we repeated the comparison at one, three, five, eight and ten discretionary slots in 2022 and 2023. In all ten year–quota combinations rank-informed covered more historical offers and placed fewer pupils in replay. The size varied sharply. At one slot, historical coverage rose by 37,861 and 42,580, while placement fell by 18,907 and 17,922. At ten slots, coverage rose by only 34 and 70 and placement fell by 48 and 51. Once all reported options fit, the policies give the same allocation by construction.

At the five-slot quota we also varied how ties are resolved within the original priority tiers in the 2022–23 replays. We drew new independent course-specific lotteries and, separately, a common applicant lottery across courses. We used 32 paired draws per ordering scheme and year, recalculating the full-list reference for each draw. The rank-informed policy raised full-partner coverage and lowered placement in every one of the 128 paired comparisons. Net placement differences ranged from −557 to −393 pupils. Historical-offer visibility stays fixed when we change only the allocation order, so it supplies no additional test in this exercise.

7 Who loses a placement?

At the level of a whole market, both studies give instances of higher coverage and fewer placements. The losses reach different people. Figure 1 partitions everyone who was placed under *A* and unplaced under *B*. We first separate people whose menu was identical. Among the rest, we ask whether their full-list partner became visible, became hidden, or had the same visibility as before.

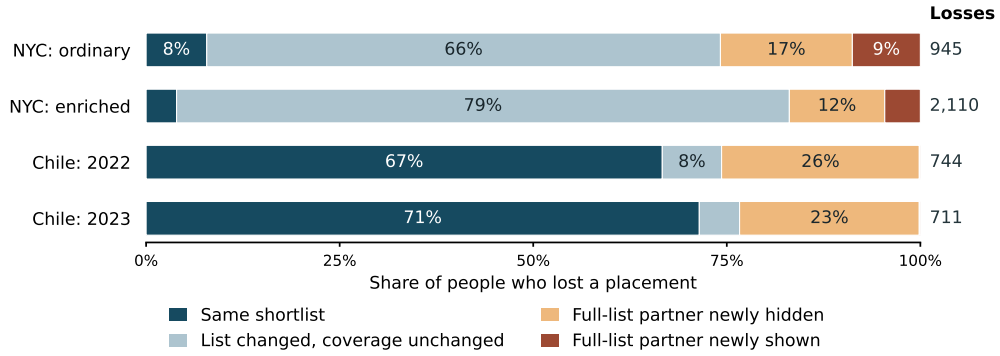


Fig. 1. What changed for people who lost a placement? Bars within each row sum to 100%. Numbers at right are raw gross losses. A changed list can retain the same partner coverage, including when the applicant had no full-list partner. The latter applicants still count in the analysis.

A change that reaches someone else's list. In the original 2022–23 Chilean cohorts, only 32,914 of 1,107,244 application records had different displayed courses under the two predictors. Of 1,455 applicants who lost a placement, 1,004 had exactly the same personal menu. The reverse happened to 406 of the 470 who gained one. Holding preferences, priorities, capacities and an applicant's own list fixed, a change in that applicant's placement can arise only through competitors' menus. These spillovers account for a net loss of 598 seats among applicants whose lists stayed fixed. Among the 32,914 applicants with edited lists, 451 lost a place (1.37%). Among the 1,074,330 with identical lists, the 1,004 losses represent 0.093%.

In the ordinary NYC markets, 945 renter instances lost a placement and 475 gained one. Of the 945 who lost, 74 saw identical menus and 83 newly saw their full-list partner. In enriched markets, 2,110 lost a placement and 1,212 gained one. Here 83 of those who lost saw identical menus and 97 newly saw their partner. Over both designs, newly seeing one's full-list partner and losing a place happened to 180 renter instances. For all 180 we found the distinct displaced-partner witnesses in Theorem 3.1 within their respective market. Of these paths 169 reached a witness after one program. Longer paths involved up to three programs.

The newly shown school offer often converted. In Chile in 2022–23, 3,750 applicants newly saw their full-list DA partner under the rank-informed predictor. Of these, 3,728 obtained that exact course in the replay and 20 obtained a different course. The remaining two went unplaced. In ordinary NYC markets 1,056 of the 1,720 newly shown renters obtained the named apartment (61.4%); in enriched markets it was 1,277 of 2,012 (63.5%). Across all applicants in each NYC design the number receiving their exact full-list partner increased by 523 and 553, respectively. Across the two school cohorts it increased by 5,049. More people therefore received their exact full-list partner even while fewer received any placement. We verified both displacement paths. Analyzing

the recorded historical offer as a separate target, 3,366 of the 3,622 applicants who newly saw that program received it in the replay. Thus a platform could document a high rate of placement at the newly shown program and still miss a loss elsewhere in the market. The witness theorem applies to the modeled full-list partner. It uses stability and cannot be applied to an arbitrary historical offer.

Feasibility and priorities. Let $v(D)$ be the largest number who can simultaneously take displayed options within capacity, and $g(D) = v(D) - |N_D|$ the feasible places unrealized by DA. The placement change is $\Delta v - \Delta g$ [Aguiar and Hong, 2026]. In ordinary NYC markets, displayed capacity fell by 3.84 percentage points and the shortfall shrank by 1.13, giving the observed 2.71-point decline. In enriched markets those changes were -12.29 and -6.89 points. The school results differ. In 2022 the display could accommodate 244 fewer applicants and the shortfall grew by 224, for a loss of 468. In 2023 those figures were 153 and 364, for a loss of 517. Each school maximum is certified by a feasible flow and a cut of equal value. This benchmark uses only displayed edges and seat counts. It relaxes priorities and the obligation to place a particular returning pupil. These identities describe two margins on the same graph. They do not isolate the causal effect of changing priorities.

8 What changes during a partial rollout?

A service might try the new rule on a fraction of applicants and use the result to judge a full switch. We simulated that decision on the original 480 rental markets and two school cohorts. Independently for each applicant, we use policy B's full menu with probability p and otherwise keep A's. We clear the entire market at each $p \in \{0, 1/4, 1/2, 3/4, 1\}$. We use 64 independent pairs of random assignments, with complementary uniforms within a pair and shared draws across uptake levels. This design gives a Monte Carlo standard error conditional on the fixed cohorts, preferences and candidate menus. For NYC we also resample whole simulation seed blocks. These experiments were specified after the original full-switch comparisons.

For any fixed reference offer, the expected number of people shown that offer is exactly the interpolation between the two endpoint counts: menu membership depends only on that person's switch. At half adoption the complementary assignments give that midpoint exactly in each pair. Placement need not follow the same line. For example, if two one-slot menus both show x before the switch and both show y afterward, either endpoint places one applicant. A mixed market with one applicant at each site places two. Reversing two initially distinct menus can bend the placement curve in the opposite direction.

Placement at half adoption exceeded the interpolation by 0.71 and 1.60 percentage points in New York. The corresponding 95% simulation-block intervals were $[0.59, 0.84]$ and $[1.40, 1.82]$, with Monte Carlo standard errors of 0.009 and 0.010 points. In Chile the midpoint deviation was about four pupils in 2022 and eight in 2023 (Monte Carlo standard errors 0.45 and 0.44 pupils, respectively). The school curves are close to linear in aggregate, yet applicants still change places. At half adoption an average of 272 pupils in 2022 and 271 in 2023 lost a place on a menu identical under both policies. The corresponding gains were 145 and 107. A nearly linear aggregate curve offers little protection to an individual family.

8.1 Which proposed lists keep a tentative allocation stable?

Suppose the platform observes an allocation M_A stable on the old lists before deciding who receives new lists. Each person can receive either their entire old row A_i or their entire proposed row B_i. Flag row i if either (i) $M_A(i)$ is a real program missing from B_i, or (ii) B_i contains a pair (i, j) that would block M_A. For the second test, i must prefer j to $M_A(i)$, and j must have a vacancy or prefer i to one of its holders. Since M_A is stable on A, only newly added options need this test. Let F be the flagged applicants. With indexed rankings, check feasibility of each old contract and

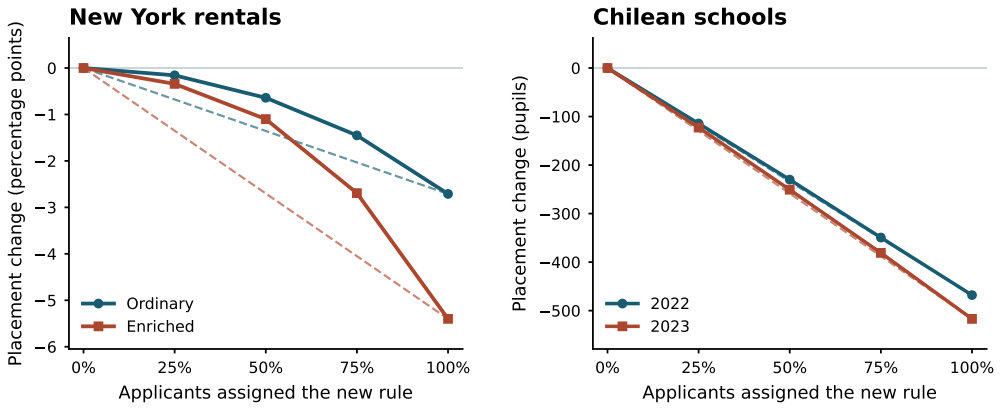


Fig. 2. Placement change from the original policy when applicants independently switch to the proposed menu. Points give the simulated mean at each uptake. Dashed lines interpolate between full adoption and the original policy. The Chile panel reports pupils within each national cohort; the NYC panel uses geographically standardized percentage points.

compute each program's occupancy and worst incumbent priority once. A scan of the proposed edges then constructs F in linear time in the explicit menu and capacity input size. This rule uses baseline assignments and the rankings relevant to these pairs. In a simulation it can use the modeled rankings; a service with incomplete information would need to elicit these comparisons.

Theorem 8.1. *If S receives its proposed rows, M_A remains stable exactly when $S \cap F = \varnothing$. On any such display every stable outcome has the original recipients, and applicant-proposing DA weakly improves every applicant's assignment relative to M_A. For an additive row objective with gain g_i from switching row i, the largest net gain that keeps M_A stable is $\sum_{i \notin F} (g_i)_+$.*

More generally, put $R = S \cap F$ and let N be any stable outcome on the mixed display. Let K_j and T_j count people in R assigned to program j under M_A and N, respectively. The gross numbers of placement losses and gains are each at most

$$\left|\{i \in R : M_A(i) \neq \perp \text{ or } N(i) \neq \perp\}\right| - \sum_j \min(K_j, T_j) \;\leq\; |R|.$$

Proof. A removed baseline partner makes M_A infeasible. A newly displayed blocker prevents its stability. Both defects depend only on the applicant's row when the allocation being tested is fixed. In their absence every old contract survives and every displayed pair passes the blocking test. Matched-set invariance and applicant optimality give the first guarantees. The objective is therefore maximized by switching precisely the unflagged rows with positive gain, with free choice on zero gains.

For the bound, remove R and its occupied seats from both allocations. In M_A all remaining assignments survive on the mixed display and no remaining pair blocks. In N stability also survives the restriction. Compare the resulting capacity vectors $c - K$ and $c - T$ using the one-seat argument of Theorem 3.1. Outside R, losses are at most $\sum_j (T_j - K_j)_+$ and gains are at most $\sum_j (K_j - T_j)_+$. Within R, losses equal $\sum_j K_j - d$ and gains equal $\sum_j T_j - d$, where d counts members matched under both allocations. Adding either margin yields the stated expression. □

The condition protects the old allocation as a stable witness. A flagged applicant can still switch without causing any placement loss. The distinction allows the rule to be checked before reallocation and limits the gains it can deliver. For independent switches with probabilities p_i, the probability that the old allocation remains stable is $\prod_{i \in F}(1 - p_i)$, and expected gross losses or gains are at most $\sum_{i \in F} p_i$. The probability factorization is related to the one-sided uncertainty result of Aziz et al. [2020]. Here program priorities stay fixed and applicant menus change.

For coverage of a fixed reference partner, $g_i = C_i(B) - C_i(A)$. The maximizing subset then gives a retrospective upper bound because the reference can be unknown at recommendation time. In the 2022–23 school cohorts, 27,851 of the 32,914 edited rows could switch together while preserving the original allocation as a stable witness. Yet only 22 of the 3,750 newly shown full-list partners occurred on those rows. Even with hindsight, this guarantee permits at most 22 more full-list hits, or 179 more historical-offer hits when that separate objective is optimized. Adopting all unflagged rows changes these two coverage counts by −3 and −46, respectively, while every applicant keeps a weakly preferred assignment. With uniform lists as the rental baseline, 1,410 of 8,309 changes in ordinary markets and 2,734 of 12,329 in enriched markets passed the screen. The respective hindsight maxima were 132 and 133 additional full-list hits. These are pooled counts over simulated renter instances. The constraint keeps all original contracts supportable; other approaches that allow contracts to move could retain more coverage.

In NYC, switching every unflagged row gained 41 full-partner hits in the ordinary draws and 22 in the enriched draws, with unchanged placement counts in both. An independent direct enumeration tested the turnover bound on 1,073,293,568 pairs of stable outcomes over ordered three-by-three displays, up to relabeling. The statement for general capacities follows from the proof above.

9 How many vacancies could one referral fill?

Once the platform has tentative offers, it knows which people and seats are still available. Suppose it can contact an unmatched applicant with one currently vacant option. A vacancy alone says little about who can use it. An available apartment may be outside the renter's budget, and an available school may be one the family never listed. In this section each referral must use a pair already present in the feasible graph. The calculation is an exploratory follow-up to both experiments.

9.1 One coordinated referral after offers

Suppose the platform now observes an assignment M_0 stable on the initial lists. Join each unmatched applicant to each mutually acceptable program with a current vacancy. These edges are hidden: if one were displayed it would block M_0. Give each program its remaining capacity and let r be the size of a maximum matching in this graph. With lists of $\min(k, d_i)$ programs, where $k \geq 1$ and d_i is feasible degree, a hidden option can always replace a displayed one. If an entitlement is protected outside the quota, the substitution is made in a discretionary slot. In both replays, every selected applicant had a full five-slot discretionary list.

Theorem 9.1. *The largest number of additions compatible with retaining every initial contract is r. One shortlist substitution per selected applicant attains it. Every stable outcome on the revised display gives each consenting applicant their named referral, even if others decline. Under applicant-proposing DA, every original recipient also receives a weakly preferred assignment.*

Proof. The new contracts in any extension must form a residual matching. Conversely, choose any such matching A. For each consenting participant substitute their assigned program for their last discretionary displayed option. Write W for the original assignment together with the accepted referrals. All new pairs in the display belong to W. On old edges the applicant weakly improved. An originally full program has the same holders. At an originally slack one, every applicant with

an old edge weakly preferred their original assignment, by stability. Hence W is stable on this display. DA weakly dominates W for applicants. All stable selections have the same matched people [Roth, 1986]. If some stable selection omitted an accepted referral e, it would keep its recipient matched even after deleting that unused edge. On that same graph $W \setminus \{e\}$ is stable by the preceding argument, since this applicant now has a subset of their original list. That would leave the person unmatched in one stable outcome only, a contradiction. □

The maximum fixes the original contracts in a stable witness. Opening new options to incumbents would be a different intervention. The empirical calculation also requires deciding which hidden options a person would actually accept. We report both a top-five rank limit and all mutually acceptable vacancies. Work on facilitating matches, adding school seats and guaranteed school referrals provides related interventions [Gajulapalli et al., 2020, Lepe, 2025, Trabelsi et al., 2025].

Table 5. Largest number of named referrals compatible with retaining all existing contracts in one stable allocation. NYC uses simulated Haiku offers. Chile uses rank-informed pooled DA offers. All counts use instances in their own row. A vacancy must be at an affordable, feasible listing or an eligible course already on the pupil's reported list.

	Vacant seats	Unplaced with an option	Any rank	Top five
NYC ordinary	7,972	757	683	204
NYC enriched	5,990	2,527	1,926	370
Chile 2022	601,565	128	127	0
Chile 2023	640,751	97	97	1

The NYC totals mask a distributional constraint. In 240 ordinary markets only 37 initially unplaced lower-income renter instances could use any vacancy. Among all ways of achieving the maximum of 683 additional placements, between 22 and 25 could go to them. In enriched markets 1,601 initially unplaced lower-income renters could use a vacancy. At a total of 1,926 additions, between 1,002 and 1,085 could go to the lower-income group. We obtained the endpoints by maximizing cardinality and then minimizing or maximizing that group's count. The enriched design changes both supply and who is drawn. The difference between the rows is a scenario comparison.

In the 2022–23 Chilean cohorts the seat totals sum all residual capacities in the course-level replay, including courses that no unmatched applicant had listed. At most 224 placements can be added over the two application years by this contract-preserving intervention, assuming full take-up of an eligible referral. One applicant could be helped within the family's top five. The other available referrals require going further down the original list. In the rank-blind comparison there were zero eligible referrals in both years. After giving each policy its own largest extension, rank-informed still placed 761 fewer applicants in total. In New York, starting from uniform lists instead gave maxima of 215 ordinary and 518 enriched placements. The number of available second-round referrals depends in part on what the first round already achieved.

A second implementation solved the school flow problems on each entire national graph and independently replayed its assignments. For every selected applicant we replaced the last discretionary option with the named listing or course and reran DA. We checked that every referral was honored, that all earlier recipients kept a weakly preferred assignment and that each list kept its original size. We preserved school continuity rights throughout. An operational service would have to elicit acceptable alternatives, observe current offers, and offer a real opportunity to decline. The counts in Table 5 measure opportunities under those assumptions. They do not measure actual take-up.

10 Implications and limits

Consider an evaluator who checks only whether the eventual offer appeared among the recommendations. In the school replays that evaluator would endorse adding rank information. Checking only applicants whose lists changed would still miss roughly seven in ten gross placement losses. A platform protecting tentative recipients faces a sharper decision. In 2022–23 it could adopt most proposed rows and still preserve the old contracts as a stable assignment. Nearly all the full-partner prediction gains, however, were on the remaining rows. A promise to support every old contract and a promise to maintain only the number placed can therefore leave very different room for redesign.

The rental experiment addresses a different social concern. A renter who cannot afford a single listing has zero hidden blocking pairs and zero chance of placement under either policy. A group can consequently score well on a hidden-options measure while having almost no access to housing. We keep these households in the denominator, report their feasible sets, and examine how much discretion the platform has once tentative offers exist. Giving lower-income renters priority among equally large referral assignments had room to move only three additional placements toward them in the ordinary draws. A service that also changes affordability or expands search would face a different feasible graph.

Limits of the evidence. The housing data identify asking rents and surveyed household characteristics. Preferences, housing-provider priorities, budgets and assignments are modeled. The budget in the prompt is correlated with income, even though group membership is absent from the prompt. Results condition on one model version, one stored response per queried renter and a small number of independent simulation blocks. In Chile the rank-informed model already has access to families' ordered applications. Its evidence concerns curation of known choices. The actual admissions system uses reserves and family contingencies beyond the pooled-priority replay. Changing what a platform displays could change what people search for, how they report, who applies, or whether they accept. Neither study estimates those responses or a causal effect of deployment. The Chilean application years may include some of the same children. We treat them as finite cohorts and report exact counts. The shared first-five benchmark uses the same rule in both settings. The site-specific recommenders address different decisions. We do not pool their effect sizes.

What a platform can audit. Record the available options and capacities, the two complete sets of proposed lists, the reference offer where one exists, and the assignment induced by each set of lists. Those records distinguish a successful recommendation to its intended recipient, a lost place on a list that never changed, and a person without any feasible alternative. The last case points to a constraint on supply. The first two call for evaluating recommendation and allocation together.

Reproducibility. The accompanying code regenerates the two studies' aggregate tables and figures. In NYC we checked 2,880 published market-by-group endpoint rows against the regenerated menus. The legacy reply cache stores listing indices without a digest of the original prompt payload. Its consistency with this reconstruction can be checked, while the exact original payload cannot be authenticated independently. In Chile we recomputed the blind, rank-informed and full-list assignments in the original two cohorts and checked all 3,321,732 applicant–policy outcomes against the archived serial solver. The 2022–23 sign criterion was recorded locally before data acquisition. The extension to 2024–25 was committed to the research repository before downloading either release. The joint transitions, alternative-baseline, rollout, sensitivity and referral analyses are follow-ups. The two additional cohorts yielded a further 2,812,215 person–policy checks. All shared school outputs are aggregates.

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